

Five-Mirror EUV Projection-Optics Simulator at 13.5 nm

Generated for the Mirrors in EUV project

September 3, 2026

1 Purpose and scope

This document describes the numerical setup, editable mirror and coating parameters, exported data, and every non-interactive plot produced by `five_mirror_EUV_projection_simulation.ipynb`. The model is intended as an inspectable research and teaching starting point for studying five reflective projection surfaces at the 13.5 nm wavelength used in extreme-ultraviolet (EUV) lithography.

The notebook deliberately separates two physical calculations:

1. **Geometrical propagation.** Optiland traces rays through five rotationally symmetric reflective aspheres and onto a wafer plane. This part determines ray intercepts, field mapping, aberration-induced spot shape, and sensitivity to curvature or aspheric figure.
2. **Multilayer reflection.** A NumPy transfer-matrix calculation evaluates a vacuum/Ru/(Si/Mo)₄₀/substrate stack. This part determines reflected power versus wavelength, incidence angle, polarization, coating thickness, pair count, and interface roughness. Coating loss weights the rays but does not alter their directions in this notebook.

2 Optical setup

2.1 Overall Structure of Mirrors

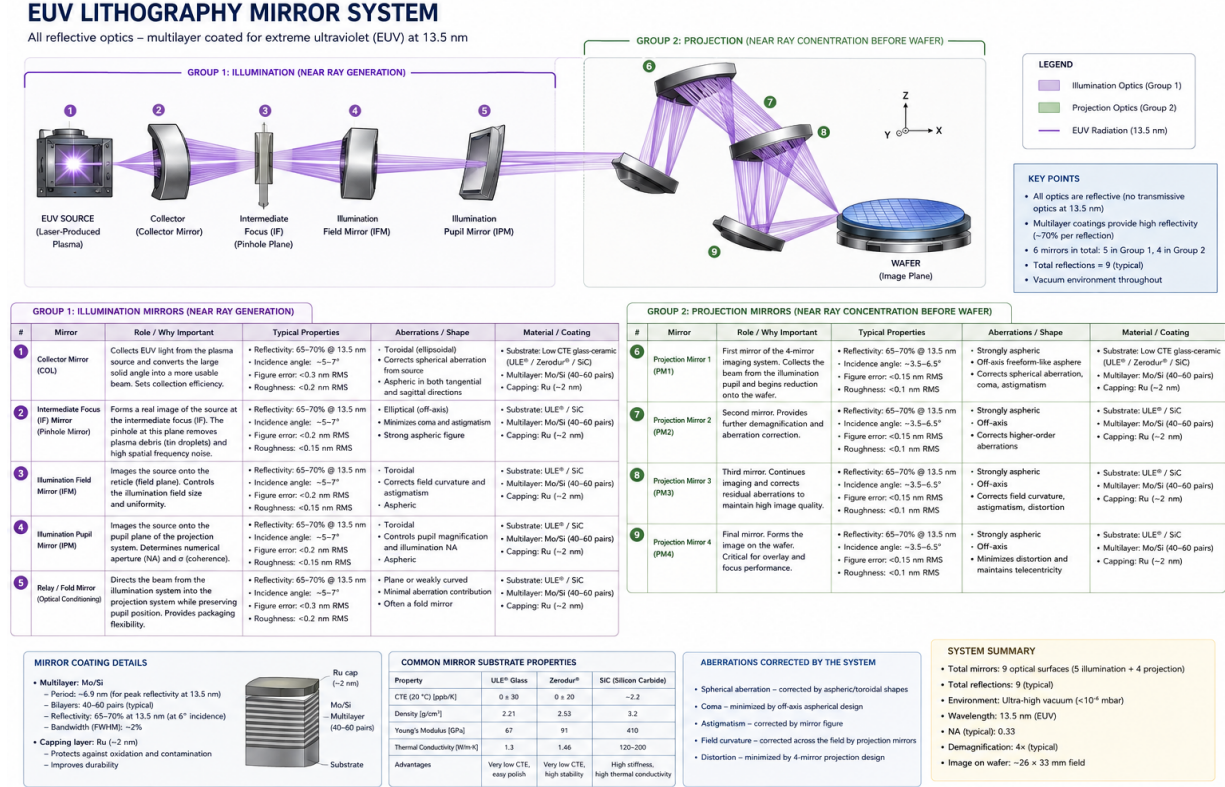


Figure 1: Setup of the two sets of Mirrors in a photo-lithography machine

2.2 Global settings and units

Table 1 summarizes the nominal global configuration. Optiland uses millimetres for spatial coordinates and micrometres for wavelength; therefore the notebook passes 13.5 nm as 0.0135 μm . The coating solver uses nanometres for wavelength, film thickness, and roughness.

Table 1: Global simulation settings.

Quantity	Nominal value	Meaning
Wavelength	13.5 nm	Monochromatic EUV ray-trace wavelength
Object distance	351.141 510 mm	Mask plane to M1 vertex
Image-space f-number	2.78	Aperture specification
Mask-field heights	−208, −211, −214 mm	Three samples across the ring field
Stop surface	M4	Physically accessible aperture stop
Random seed	7	Repeatable random-pupil spot sampling

The system is surface-sequential. A surface's “thickness to next” is the signed axial distance from that surface vertex to the next surface vertex in the current propagation direction. Because a mirror reverses propagation, the signs alternate through the folded sequence. These negative spacings are therefore intentional; replacing them by absolute values does not merely move a component but changes the optical path topology.

The mask samples lie near the centre of the disclosed annular field. In calls to `system.trace`, $H_y = -1, 0, +1$ are *normalized field coordinates*; they select the two field endpoints and the central

field. They are not millimetre coordinates. The corresponding physical mask heights are specified by `FIELD_Y_MM`, while the actual image-plane intercepts are exported in millimetres.

2.3 Five-mirror prescription

The default geometry is based on the public five-mirror, 4:1 reduction, NA 0.18 prescription in US6072852A. Its power sequence is concave–convex–concave–convex–concave. Table 2 lists the values entered in the notebook. Radius is the vertex radius of the base conic. A negative radius follows the patent/Optiland sequential sign convention; it should not be interpreted without the signed spacing and current propagation direction.

Table 2: Nominal mirror settings. AOI is measured from the surface normal.

Mirror	Radius (mm)	To next (mm)	K	AOI (deg)	AOI half-span (deg)	Role
M1	-1277.318653	-292.381122	1.626111	11.34	2.0	Concave power
M2	-417.634186	504.794891	0.339465	7.72	2.0	Convex power
M3	-757.329052	-496.445439	0.026174	5.28	2.0	Concave power
M4	-448.096932	193.582965	2.592761	15.52	3.0	Convex power; stop
M5	-392.026187	-260.692804	0.346595	7.76	2.0	Concave power

The AOI half-span is used only by the approximate pupil-apodization plot. It represents a linear engineering estimate of angle variation across each beam footprint, not a recovered local-angle map from the patent. Changing this number modifies the pupil-intensity estimate but not the Optiland ray path.

Every default surface is an even asphere with sag

$$z(r) = \frac{cr^2}{1 + \sqrt{1 - (1 + K)c^2r^2}} + Ar^4 + Br^6 + Cr^8 + Dr^{10}, \quad c = \frac{1}{R}. \quad (1)$$

Optiland stores an even-asphere coefficient vector beginning at r^2 , so the notebook passes $[0, A, B, C, D]$. Table 3 lists the deformation coefficients. Their dimensions are those needed for z and r in millimetres: A has units mm^{-3} , B mm^{-5} , C mm^{-7} , and D mm^{-9} .

Table 3: Even-asphere deformation coefficients in Equation (1).

Mirror	A	B	C	D
M1	0	3.498200×10^{-16}	$-7.101160 \times 10^{-22}$	0
M2	0	5.759030×10^{-15}	9.512560×10^{-20}	0
M3	0	1.143970×10^{-17}	7.746320×10^{-23}	0
M4	0	$-9.009250 \times 10^{-15}$	$-2.722770 \times 10^{-19}$	0
M5	0	$-9.701720 \times 10^{-16}$	$-1.567560 \times 10^{-20}$	0

The editable mirror dictionaries also expose clear-aperture radius, x/y decenter, and x/y tilt. Aperture radius is left as **None** by default, allowing Optiland to establish the working semi-apertures from the f-number and stop. Tilts are in radians; radii, spacings, and decentres are in millimetres.

3 Mo/Si multilayer and Ru cap

3.1 Layer sequence

Each mirror has an independent coating dictionary. From the incident vacuum toward the substrate, the nominal stack is

$$\text{vacuum/Ru}(1.50 \text{ nm})/[\text{Si}(4.15 \text{ nm})/\text{Mo}(2.80 \text{ nm})]_{40}/\text{substrate}. \quad (2)$$

The nominal bilayer period is therefore 6.95 nm. The dictionary permits different Mo, Si, and Ru thicknesses on every mirror, different pair counts, linear depth grading, and arbitrary individual-layer overrides. Individual layers are named from the vacuum side, for example `Si_01`, `Mo_01`, and `Mo_40`. The files `data/layer_stack_M1.csv` through `data/layer_stack_M5.csv` record the fully expanded stack actually used.

The reference complex indices at 13.5 nm are shown in Table 4. The notebook uses the passive convention $\tilde{n} = n + ik$ with $k > 0$ and a propagation phase that decays in absorbing media.

Table 4: Reference complex refractive indices at 13.5 nm.

Material	n	k
Vacuum	1.000000000	0
Ru	0.886360034	0.017064894
Si	0.999002305	0.001826494
Mo	0.923793525	0.006435425
Effective substrate	0.973713000	0.012976400

These indices are held constant over the narrow 12.5 nm to 14.5 nm plotting interval. Consequently, the plotted wavelength sensitivity captures the interference-phase and thickness effects cleanly, but it is not a substitute for a measured dispersive $n(\lambda) + ik(\lambda)$ dataset when absolute broadband accuracy is required.

3.2 Reflectivity calculation

For incident angle θ_0 , Snell’s law is applied in complex form to each layer j ,

$$\cos \theta_j = \sqrt{1 - \left(\frac{n_0 \sin \theta_0}{n_j} \right)^2}, \quad k_{z,j} = \frac{2\pi}{\lambda} n_j \cos \theta_j, \quad (3)$$

with the square-root branch selected to give passive propagation. The optical admittances are $q_j^{(s)} = n_j \cos \theta_j$ and $q_j^{(p)} = \cos \theta_j / n_j$. The interface amplitude is

$$r_{ij} = \frac{q_i - q_j}{q_i + q_j}. \quad (4)$$

The recursion combines every interface amplitude with the round-trip layer phase $\exp(2ik_{z,j}d_j)$. A compact Nevot–Croce-like factor

$$\exp[-2|k_{z,i}k_{z,j}|\sigma^2] \quad (5)$$

reduces each interface amplitude for nominal RMS roughness $\sigma = 0.25$ nm. This term is intentionally approximate: it does not distinguish topographic roughness from interdiffusion or silicide formation. The unpolarized power reflectivity is $R = (|r_s|^2 + |r_p|^2)/2$.

4 Plots and Interpretations

4.1 Mirror reflectivity and cumulative intensity

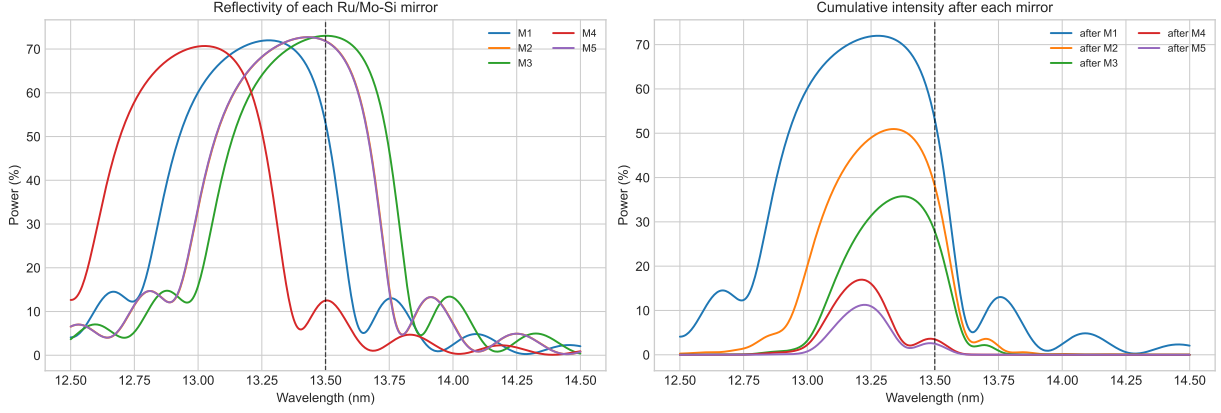


Figure 2: Individual multilayer reflectivity (left) and cumulative five-mirror intensity (right).

Left-panel axes. The horizontal axis is vacuum wavelength in nanometres, from 12.5 nm to 14.5 nm. The dashed vertical line marks the operating wavelength, 13.5 nm. The vertical axis is reflected optical power as a percentage of the power incident on the particular mirror. Each coloured curve uses that mirror’s own nominal AOI and coating dictionary. Curves can differ even when their physical layer thicknesses are identical because the normal component of wavevector and hence the Bragg condition change with AOI.

Right-panel axes. The horizontal axis and dashed reference line have the same meanings. The vertical axis is power remaining relative to the power entering M1. The curve labelled “after Mn ” is

$$I_n(\lambda)/I_0 = \prod_{j=1}^n R_j(\lambda). \quad (6)$$

The curves must therefore move downward as more mirrors are included. This multiplicative loss is why a few percentage points of reflectivity improvement per surface can materially change total scanner throughput. The exported `data/reflectivity_spectrum.csv` stores wavelength, every individual reflectivity as a fraction, and every cumulative fraction. Percent values in the plot are 100 times those fractions.

At the dashed 13.5 nm line, the present default calculation gives $R_1 = 53.10\%$, $R_2 = 71.84\%$, $R_3 = 73.03\%$, $R_4 = 12.50\%$, and $R_5 = 71.79\%$. Their product is only 2.501%. M4 is the dominant loss in this particular baseline because it has the largest nominal AOI (15.52°) while all five mirrors initially share one ungraded stack. This should be interpreted as a coating-tuning result, not as an intrinsic inability of a five-mirror system to transmit more power. A per-mirror stack optimized for M4’s AOI would shift its Bragg maximum toward 13.5 nm and materially improve the product.

Changing Mo or Si thickness primarily translates and reshapes the Bragg peak; changing pair count changes peak build-up and bandwidth; increasing roughness usually lowers peak reflectivity; and changing the Ru cap affects both absorption and the phase presented to the periodic stack.

4.2 Estimated coating apodization across the pupil

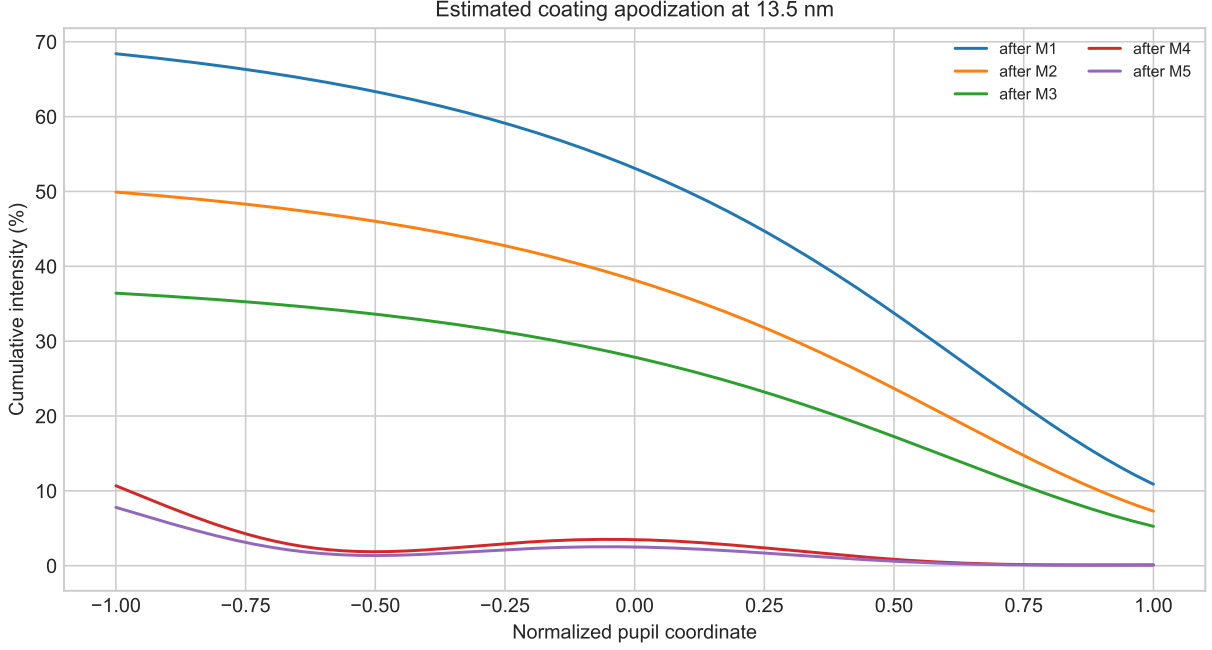


Figure 3: Estimated cumulative coating intensity across a normalized pupil diameter at 13.5 nm.

The horizontal axis is a dimensionless normalized pupil coordinate p . $p = -1$ and $p = +1$ are opposite edges of the sampled pupil diameter, while $p = 0$ is its centre. It is not a millimetre coordinate on any one mirror. For mirror j , the notebook assigns

$$\theta_j(p) = \bar{\theta}_j + \Delta\theta_j p, \quad (7)$$

where $\bar{\theta}_j$ is the nominal AOI and $\Delta\theta_j$ is the editable half-span in Table 2. The vertical axis is cumulative power relative to the power entering M1, expressed in percent. Each curve again adds one mirror to the product, now evaluated at a pupil-dependent AOI.

Slope or asymmetry in these curves represents coating-induced pupil apodization: different parts of the pupil reach the wafer with different amplitudes. Such amplitude variation can alter the effective pupil and, in a wave-optical model, the point-spread function and imaging contrast. The plot is a sensitivity estimate, not a rigorous coating map, because the assumed AOI variation is linear and one-dimensional. The CSV `data/pupil_apodization.csv` includes p , local AOI, individual reflectivity, and cumulative intensity after each mirror.

The asymmetry is large for the current common-stack example. The final five-mirror fraction is approximately 7.79% at $p = -1$, 2.501% at $p = 0$, and 0.0717% at $p = +1$. The collapse toward $p = +1$ follows the assumed increase in AOI and the strong off-Bragg response of M4. Because the AOI half-spans are illustrative rather than extracted from local ray normals, these endpoint values are best used to compare coating choices, not as a certified scanner pupil-uniformity prediction.

4.3 Optiland meridional ray diagram

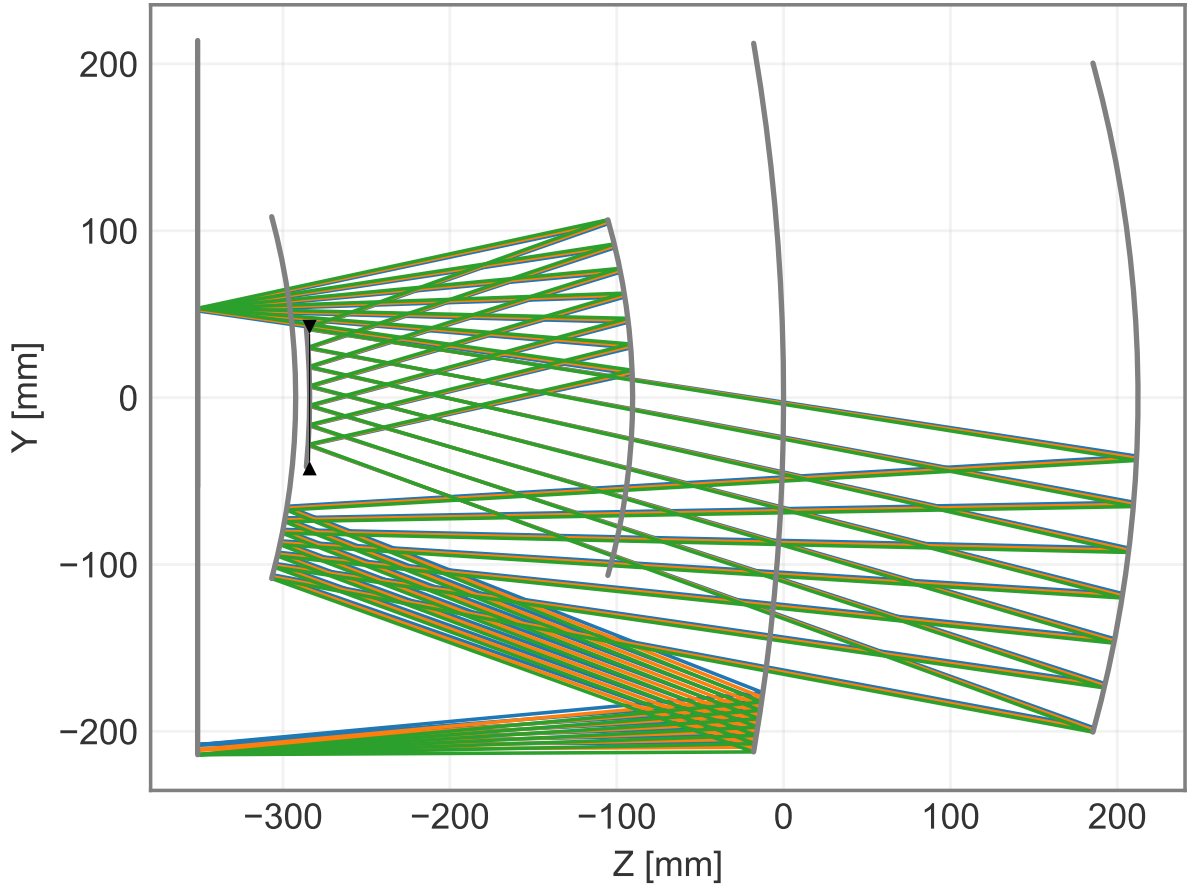


Figure 4: Sequential Optiland ray diagram for the five reflective aspheres and three mask-field samples at 13.5 nm.

The horizontal axis is Optiland global Z in millimetres. It is the meridional layout direction and should not be confused with propagation distance accumulated along a ray. Because the path folds at every reflection, a later surface in sequential order can have a smaller global Z coordinate. The vertical axis is global meridional height Y in millimetres. It shows the off-axis ring field, mirror footprints, and convergence or divergence of the sampled ray bundles. The plotted lines are geometrical rays; their separation is a pupil-sampling choice and is not an intensity scale. Surface arcs show the local conic/aspheric profiles over their working apertures. M4 is the stop and therefore determines the accepted bundle in conjunction with the image-space f-number.

This plot should be read topologically: follow each ray in surface order mask–M1–M2–M3–M4–M5–wafer rather than assuming monotonically increasing Z . A blank or discontinuous bundle after parameter editing normally means that rays miss a surface, encounter an invalid sag domain, or are vignetted by an imposed aperture. Restore the signed spacings before diagnosing smaller curvature changes.

4.4 Final wafer-plane ray shape

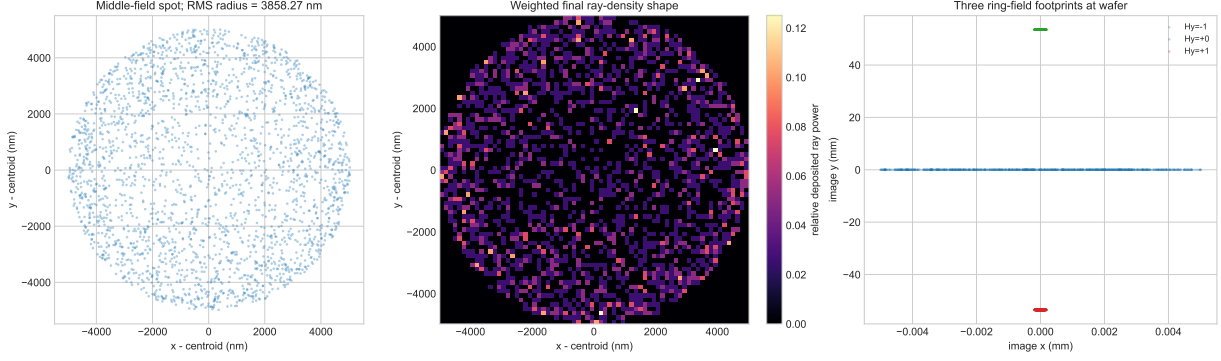


Figure 5: Centred geometrical spot, weighted ray-density map, and three-field wafer footprints.

Figure 5 contains three related views.

Left panel: centred middle-field spot. Both axes are image-plane displacement from the weighted middle-field centroid in nanometres. One plotted point is one surviving geometrical ray. Equal scales on the two axes preserve spot shape. The title reports the weighted RMS radius

$$r_{\text{RMS}} = \sqrt{\frac{\sum_i w_i [(x_i - \bar{x})^2 + (y_i - \bar{y})^2]}{\sum_i w_i}}, \quad (8)$$

where w_i includes Optiland ray intensity and the five-mirror coating throughput. Centring removes field position so that coma, astigmatism, and other shape changes are visible. It also hides absolute distortion, which is why the right panel is required.

Middle panel: weighted ray-density shape. The horizontal and vertical axes are the same centred nanometre coordinates. Colour represents the sum of relative deposited ray power within each two-dimensional histogram bin. A brighter bin contains more weighted rays; the colour scale is relative, not calibrated irradiance in W m^{-2} . Monte Carlo granularity decreases as the number of traced rays is increased.

Right panel: three physical field footprints. The axes are absolute image-plane x and y in millimetres. Colours identify normalized fields $H_y = -1, 0, +1$. Separation of the three groups shows field mapping and magnification; displacement from ideal field positions indicates distortion. The millimetre scale is much larger than the nanometre centred spot scale in the other panels, so individual aberration shapes can appear compressed here.

Raw middle-field intercepts are in `data/middle_field_spot.csv`; all three fields are in `data/three_field_footprints.csv`; and throughput, valid-ray count, and RMS radius are in `data/wafer_spot_metrics.csv`. Coordinates in the CSV remain in millimetres so no precision is lost through display-unit conversion.

For the recorded run, all 2500 middle-field rays reached the image plane. The reported geometrical RMS radius is 3858.27 nm, or about $3.86 \mu\text{m}$, and the coating weight is 2.501%. The three field centroids lie near image $y = +53.5003, 0$, and -53.5003 mm for $H_y = -1, 0, +1$, respectively, consistent with the inverted ring-field mapping. The centre field has a small mean x offset of roughly -1.16×10^{-4} mm in the finite Monte Carlo sample. These numbers describe this public starting prescription and sampling scheme; the micrometre-scale geometrical spot is not a claim of production-lithography performance.

4.5 M3 curvature sensitivity

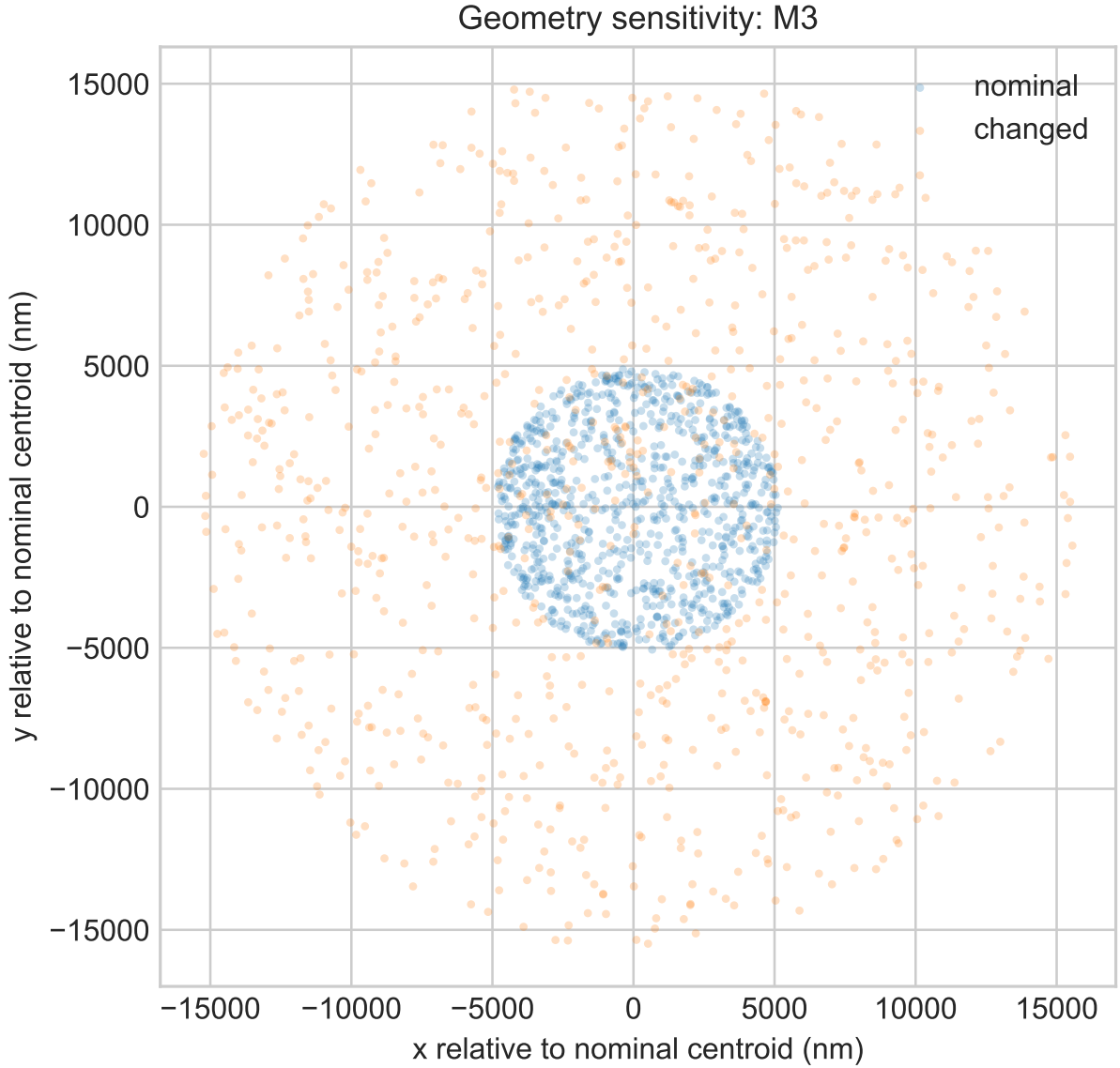


Figure 6: Middle-field spot sensitivity to a $+0.02\%$ change in the M3 vertex radius.

The demonstration multiplies the M3 radius by 1.0002 while retaining its conic, aspheric coefficients, spacing, and all other mirrors. Both axes are nanometres relative to the *nominal* spot centroid. The nominal and changed clouds share one reference origin; therefore a changed-cloud offset is a real image-centroid shift rather than an artefact removed by independently centring the two cases. A change in cloud width or shape indicates altered aberration, while a mostly rigid translation indicates image displacement.

The comparison is a one-variable sensitivity experiment, not a tolerance allocation. A realistic tolerance study would vary radius, spacing, decenter, tilt, low- and mid-spatial-frequency figure terms, and compensator settings, then report distributions over many trials. The plotted points and weights are exported to `data/geometry_sensitivity_m3.csv`. Calling the helper for another mirror creates a correspondingly named CSV and figure, but those additional user-generated plots are not automatically included in this README.

Numerically, the demonstration changes M3 from approximately -757.329052 mm to -757.480518 mm. The changed centroid moves by 244.59 nm. With both cases measured about the nominal

centroid, the weighted RMS radius grows from approximately $3.84\,\mu\text{m}$ to $10.68\,\mu\text{m}$. The orange cloud is therefore not merely translated: its much larger extent shows a strong degradation of the nominal aberration balance from a seemingly small 0.02% radius perturbation.